



Term Evaluation Theory (Even) Semester Regular Examination February 2026

Roll no.....

Name of the Course: BCA
Semester: VI
Name of the Paper: Cyber Security
Paper Code: TBC603(3)
Time: 1.5 hour

Maximum Marks: 50

Note:

- (i) Answer all the questions by choosing any one of the sub-questions
- (ii) Each question carries 10 marks.

- Q1. (10 Marks)
a. Explain the need for Cyber Security. Discuss the major factors that have increased cyber threats in modern computing environments. Illustrate the concept using the CIA Triad. (CO1)
OR
b. Define Cyber Security. Explain Vulnerability, Threat, and Harmful Acts with suitable real-world examples. (CO1)
- Q2. (10 Marks)
a. Describe different types of malwares in Cyber Security. How do they affect computer systems? (CO1)
OR
b. Explain various Cyber Attacks, Services, and Security Mechanisms. Also discuss the role of computer criminals in cyber-crimes. (CO2)
- Q3. (10 Marks)
a. What is Cyber Crime? Differentiate between Cyber Warfare and Cyber Espionage with suitable examples. (CO2)
OR
b. Describe the design principles of Block Ciphers. Explain Simplified DES (S-DES) with its basic working steps. Also discuss various Block Cipher Modes of Operation with suitable examples. (CO2)
- Q4. (10 Marks)
a. Explain the National Cyber Security Policy of India. Write its objectives and key features. (CO1)
OR
b. Explain layers of security in Cyber Security. How does a multi-layered security approach protect systems from attacks? (CO2)
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- Q5. (10 Marks)
a. Explain the Conventional Encryption Model with a neat diagram. Also explain the role of encryption in cyber security. (CO3)
OR
b. Write short notes on any two: (CO1)
(i) Steganography
(ii) Block Ciphers vs Stream Ciphers
(iii) Differential & Linear Cryptanalysis